Midterm API Project - COMP229

Project Overview

Your task for the COMP229 midterm is to wire up a RESTful API that manages a collection of equipment. An Express server is already configured to run on port `3000`, and a starter collection of equipment data is provided - you just need to implement each endpoint.

Duration: 120 minutes.

Download the instructions PDF.

Endpoints

1. **GET /api/equipment** - Retrieve the full list of equipment.
2. **GET /api/equipment/filter?category=[category name]** - Retrieve equipment whose category matches the query.
3. **GET /api/equipment/:id** - Retrieve an equipment item by its array index (ID).
4. **POST /api/equipment** - Add a new equipment item to the collection.
5. **PUT /api/equipment/:id** - Update an equipment item by its array index (ID).
6. **DELETE /api/equipment/:id** - Remove an equipment item by its array index (ID).

The starter data lives in `server.js`, but **you must complete the logic** for every endpoint above. Array indexes start at **0**: `/api/equipment/0` refers to the first item.

Array Example

Below is an example of the object structure you will manipulate. **An array containing five equipment items is already provided in the code.**

Array Example - Equipment

```
let equipment = [
  {
    name: 'USB Microphone', category: 'audio',
    brand: 'AudioLab', year: 2021, quantity: 5
  },
  {
    name: 'LED Panel', category: 'lighting',
    brand: 'BrightWorks', year: 2022, quantity: 4
  }
];
```

Array Example

Personalized Requirements

Tailor the Collection Using Your Student ID

Add exactly two equipment items to the starter array based on the last digit of your student ID:

- **0-1:** Add 2 equipment items released between **1990-1999**
- **2-3:** Add 2 equipment items released between **2000-2009**
- **4-5:** Add 2 equipment items released between **2010-2019**
- **6-7:** Add 2 equipment items released between **2020-2026**
- **8-9:** Add 2 **portable** equipment items (any year)

Example: Student ID `301234567` ends in 7, so add two equipment items released between 2020-2026.

Format for Your Additional Equipment

```
{
  name: 'Equipment Name',
  category: 'Category',
  brand: 'Brand Name',
  year: 2024,
  quantity: 2
}
```

Screenshot Requirement

Capture at least one API response that clearly displays the entire personalized collection (starter equipment + your two additions = 7 total).

Setup Instructions

1 **Open the starter project**

Choose the option that matches how you obtained the starter:

- **Already downloaded or extracted the starter?** Extract the ZIP if needed, then open the project folder in VS Code. Choose the folder containing `package.json` and `server.js`. Continue to step 2.
- **Need the starter?** Download the starter ZIP, extract it, and open the folder containing `package.json` and `server.js` in VS Code. Continue to step 2.
- **Prefer to clone with Git?** Run the commands below, then open the cloned folder in VS Code. Continue to step 2.

```
git clone https://github.com/CENT-COL/COMP229-F26-MIDTERM.git
cd COMP229-F26-MIDTERM
```

2 **Open a terminal in the project folder**

In VS Code, select **Terminal > New Terminal**. Make sure the terminal is in the folder containing `package.json` and `server.js`, whether you downloaded or cloned the starter.

3 **Install dependencies**

```
npm install
```

4 **Run the server**

```
npm start
```

The server listens on port `3000`. Stop it with Ctrl+C and restart after editing.

5 **Test the API**

Use Postman (desktop or VS Code extension), Thunder Client, Insomnia, or a similar tool. For each endpoint:

- Make the request using the correct method and URL.
- Include the request body for POST and PUT calls, using raw JSON.
- Verify the response payload and HTTP status code.

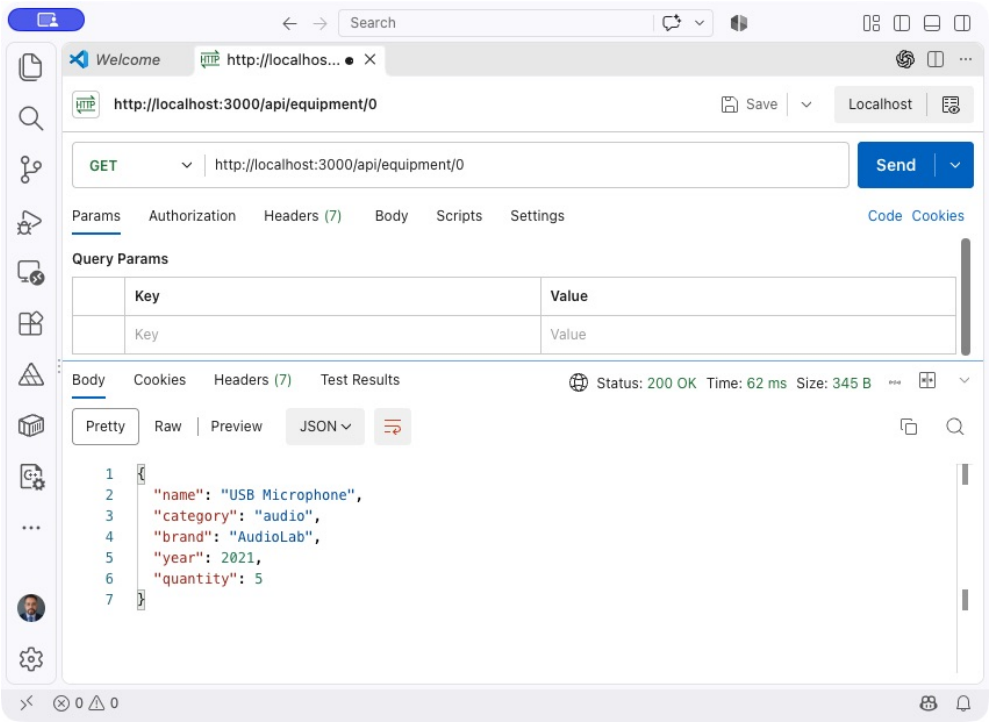
Submission Requirements

1 **Screenshots** Provide clear evidence of testing for **all six endpoints**. Every screenshot must show:

- Request URL and method
- Request body when applicable
- Successful response with status code
- At least one screenshot highlighting your complete 7-item equipment collection

Filename convention: `[StudentID]_[LastName]_[EndpointNumber]_[Method].png` Example: `301234567_Smith_01_GET_ALL.png`

Need a reference? The screenshot below shows the expected Postman layout.



Postman Example

2 **Code Submission**

- Zip your project files, excluding `node_modules` and `.git`.
- Share the GitHub repository link containing your implementation.
- Keep screenshots organized and easy to review.
- Confirm your personalized equipment appears in both code and responses.

3 **Reflection (optional, but recommended)** Summarize in a short paragraph how you validated each endpoint and any edge cases considered.

Academic Integrity Notice

Warning: Submissions are reviewed for

- Personalized additions that match your student ID rule set
- Proper screenshot naming conventions
- Individual equipment selections across submissions

Suspected irregularities will be investigated according to academic integrity policies.

Good Luck!

Write clean, well-structured code, validate inputs, and handle errors gracefully. You've got this!